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Isothermal Microcalorimetry as a Tool to Study Solid-Electrolyte Interphase Formation in Lithium-Ion Cells

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Isothermal microcalorimetry can be used in conjunction with electrochemical measurements to study solid-electrolyte interphase (SEI) formation reactions as they occur in a Li-ion cell. The heat flow was measured in wound cells that contained no electrolyte additives and in cells prepared with four additives that are known to produce an SEI at the negative electrode surface: vinylene carbonate (VC), fluoroethylene carbonate (FEC), pyridine boron trifluoride (PBF), and prop-1-ene-1,3-sultone (PES). For VC, two distinct features in the differential capacity (dQ/dV vs. Q) plot that align with overlapping peaks in the heat flow plot do not agree with a simple one-electron reduction followed by anionic polymerization. For FEC, three distinct differential capacity and calorimetric peaks are observed. Heat flow measurements at multiple PBF concentrations show that PBF reduction does not significantly affect the reduction of EC at higher cell voltage. The total heat flow during SEI formation in PBF- and PES-containing cells match the calculated energies in recently published reaction pathways. It is concluded that IMC may be used to study the underlying chemistry of SEI formation, especially when paired with computational studies.

Introduction

When a lithium-ion cell is charged for the first time, a solid-electrolyte interphase (SEI) is formed at the negative and, at higher cell voltages, the positive electrode-solution interfaces.¹ Moreover, the lifetime and performance of a cell is significantly affected by the chemical composition and morphology of these SEI layers, even after several years have passed since their initial formation. SEI-forming electrolyte additives are widely adopted to modify these surface layers, which leads to improved cell performance and extended calendar lifetimes. However, the underlying reaction pathways for SEI formation remain uncertain for most additives, in part due to the significant technical challenges involved in studying these processes in detail. Clearly, new methods for studying the formation of the SEI as it occurs would benefit the development of improved lithium-ion cell chemistries.

With this motivation, this article demonstrates a new use for isothermal microcalorimetry (IMC) in lithium-ion battery research. IMC is an analytical technique in which the heat flow is measured from a system that is maintained at a constant temperature. This method may be combined with electrochemical measurements to study the heat flow in half cells for various positive^{2–11} and negative^{5,9,12,13} electrode materials. It may also be used to study the heat flow during

cycling in full cells,^{9,14–17} and it has provided new insight into the degradation of the active electrode materials.^{15,17} More recently, IMC has been applied to full cells to measure the onset of lithium plating on the negative electrode,¹⁸ small heat flows from parasitic reactions, such as decomposition of the electrolyte components,^{19,20} and the effects of electrolyte additives on these parasitic processes.^{21–24} Recently, IMC was applied to study the reactions involved in SEI formation for cells prepared with pyridine boron trifluoride (PBF), a promising electrolyte additive.²⁵ It was therefore concluded that further work was merited to consider what sort of new insights IMC measurements may offer into SEI formation.

On its own, the heat flow of SEI formation is not obviously informative. However, if one could predict the heat flow for possible reaction mechanisms, then one could compare these predictions with the experimental heat flow to evaluate proposed pathways. Given the use of computational studies to develop new understanding of solvation, electrolyte decomposition and SEI formation,^{25–36} the use of DFT to calculate reaction energies seems a likely route to predict the heat flow of SEI formation. Therefore, this work tests the hypothesis that the pairing of IMC with computational studies can be used to investigate the underlying chemistry of SEI formation. In the present work, the heat flow has been measured during SEI formation in Li(Ni_{0.42}Mn_{0.42}Co_{0.16})O₂/graphite (NMC442) cells prepared with four common additives that are known to form an SEI at the negative electrode. In particular, vinylene carbonate (VC) and fluoroethylene carbonate (FEC) were chosen because they find extensive use in lithium-ion cells with graphite and silicon anode materials, respectively.^{1,37–40} PBF and prop-1-ene-1,3-sultone (PES) were

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chosen because there have recently been detailed pathways, with DFT energies, reported for these two additives.^{25,30}

Experimental

Electrolyte Solutions

All solutions in this work used a 3:7 solvent blend, by mass, of ethylene carbonate (EC) and ethyl methyl carbonate (EMC), as received from BASF (water content < 20 ppm). The electrolyte salt, 1 mol L⁻¹ LiPF₆ (BASF, 99.94%, water content < 14 ppm), was added no more than 6 hours before cell filling. Vinylene carbonate (VC, BASF, 99.97%), fluoroethylene carbonate (FEC, BASF, 99.94%), pyridine boron trifluoride (PBF), and prop-1-ene-1,3-sultone (PES, Lianchuang Medicinal Chemistry Co., Ltd., China, 98.2%), were added to this control solution in the specified concentration. The synthesis of PBF and the evaluation of its purity by ¹H NMR (Bruker Avance 300) has been described previously.⁴¹

Pouch Cells

Dry NMC442 pouch cells (C ~ 220 mAh) were received from LiFun Technology (Xinma Industry Zone, Tianyuan District, Zhuzhou, Hunan, China). All cells were vacuum sealed without electrolyte in China and then shipped to our laboratory in Canada. Before electrolyte filling, the cells were cut just below the heat seal and dried under vacuum at 80°C for 14 h. The cells were transferred directly into an argon-filled glove box, where they were filled with 0.9 g of electrolyte solution. The cells were then vacuum-sealed at a gauge pressure of -90 kPa using a compact vacuum sealer (MSK-115A, MTI Corp.).

Isothermal Microcalorimetry (IMC)

Heat flow measurements using IMC were performed using a TAM III calorimeter (TA Instruments) at a temperature of 40.0°C. The specifics of the instrument calibration and operation, background information, and methods are discussed in detail by Krause et al.²⁰ The heat flow measurement uncertainty was < ±1.0 μW, the noise level of the instrument is about 10 nW and the baseline drift from 0.0 μW was less than 500 nW over the time frame of the experiments conducted here.

Cells were connected to a Maccor 4000 automated test system (Maccor Inc.) and inserted into the TAM III. Cells were held at 1.5 V for 24 hours in the TAM III to ensure proper wetting and stable heat flow readings before the beginning of each formation. Cells were then charged at 2 mA (C/110) from 1.5 V to 3.5 V. These are typical formation conditions for experimental and industrial lithium-ion cells, with the exception of the slow charging rate, where C/20 (22 mA for the cells in this work) or more rapid rates are more common.³⁸ The lower charging current was chosen to limit the heat flow to within the safe operating range of the calorimeter and so that heat flows due to cell polarization (proportional to *i*²) could be neglected.

Density Functional Theory (DFT)

DFT calculations were performed using Gaussian09 (revision d01) and the B3LYP/6-311++G(d,p) level of theory. Solvation effects were modelled implicitly using the IEFPCM-UFF method (ε = 20).^{28,30} A full discussion of the calculation methods can be found elsewhere.³⁰

Results and discussion

In control cells, which do not contain any additives, the solvent component ethylene carbonate (EC, Figure 1a) is electrochemically reduced at the negative electrode surface and forms a passive SEI.^{42,43} The observed voltage plateau in Figure 2a, at ~2.8 V, is thusly assigned to the reduction of EC, which is thought to primarily produce lithium ethylene dicarbonate (LEDC) and ethylene gas. Although the exact reaction pathway has not yet been determined,⁴² Figure 2b shows that a maximum heat flow of 3.93 mW is measured during this reaction. At the applied current of 2 mA, this value corresponds to ~1.97 V (*i.e.*, 1.97 eV/electron). It is prudent to first consider how much of this heat flow originates from the SEI formation reactions. Other contributing factors to heat flow, besides formation processes, include cell polarization (∝ *i*²), reversible entropy terms related to the electrode materials (∝ *I*) and parasitic reactions such as electrolyte oxidation (∝ *I*⁰).²⁰ At low potential the latter may be considered negligible, as may the cell polarization term at the low rates used here (< 1 μW at 2 mA).¹⁹ The contribution of the entropy term from the negative electrode may be ignored, because little or no lithiation occurs during SEI formation. The contribution of the entropy term from the positive electrode may be estimated from the mean field theory approach.⁴⁴ In this work, this entropy term is < 250 uW after 2 mA h charge has passed, and the average entropy term from 2 – 10 mA h is ~200 uW. Therefore, most of the experimentally measured heat flow is attributable to the SEI formation reactions. Future studies could use computational chemistry to map out possible reaction pathways and utilize this data to evaluate their results. The heat flow decreases to ~0.5 mW (0.25 V) after SEI formation is complete and as the cell begins to charge. Next, cells were prepared with electrolyte additives and the experiment was repeated. Four additives were chosen that are known to form an SEI at the negative electrode: vinylene carbonate (VC, Figure 1b),^{45–48} fluoroethylene carbonate (FEC, Figure 1c),^{40,49–51} prop-1-ene-1,3-sultone (PES, Figure 1d),^{30,47,48,52–55} and pyridine boron trifluoride (PBF, Figure 1e).^{25,41,56} In order to demonstrate the capabilities of this technique, the differences between SEI formation in VC- and FEC-containing cells will first be discussed. There is good evidence that both additives form a passive SEI on the negative electrode surface via polymerization, although details, such as how initiation occurs, remain uncertain.^{1,40,46,49,57–59} Therefore, these additives make an interesting pair with which to begin the discussion of how IMC may be applied to study SEI formation.

It is generally reported that the electrochemical reduction of VC occurs at lower full cell voltage than EC and produces radical anions.⁵⁷ Density functional theory (DFT) calculations have demonstrated that these anions may initiate VC polymerization, although this scheme has not been thoroughly tested.⁶⁰ For FEC, it has been suggested that lithium alkoxide species are formed by reaction of the solvent and react with FEC to produce VC and LiF.^{49,61} The VC is then suggested to polymerize in the same fashion as if it were added directly to the cell. However, this is just one theory out of several mechanisms for SEI formation from FEC that have been suggested.^{40,62–64} It is worth noting that the reactions on graphite surfaces may not be the same as those that occur on Si-based negative electrodes.

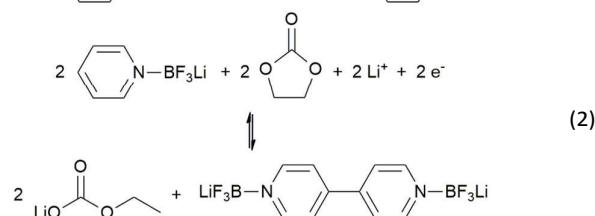
From the results shown in Figure 2, the heat flow corresponding to the various potential plateaus may be evaluated. However, it is difficult to see small plateaus and inflections in the voltage plots (Figure 2a). For this reason, differential capacity plots (dQ/dV vs E_{cell}) are a common way to present galvanostatic data. However, when these plots were aligned with the corresponding heat flow data (dq/dt vs. E_{cell}), the results were visually misleading, because of the dependence on the time constant of the calorimeter ($\tau \approx 5$ min). That is, the heat flow should be plotted against an x-axis that is proportional to the experimental time, such as the cell capacity (Q). Figure 3 shows the overlay of the heat flow and the differential capacity plotted using the cell capacity as the x-axis. In this graph, the area of dQ/dV vs. Q holds no clear physical meaning. However, this plot allows easier visual comparison of electrochemical features with their corresponding heat flow than Figure 2. For example, the VC-containing cell displays peaks at ~ 0.003 and ~ 0.005 Ah that align with a feature in the heat flow plot (Figure 3b). The maximum for this feature is at 3.78 mW (1.89 V). One may then refer back to Figure 2a for the corresponding cell voltage, which is ~ 2.65 V. The IMC results also reveal an unusual feature that precedes the electrochemical peaks. The maximum heat flow of this additional peak is 3.72 mW (1.86 V). Although the presence of two electrochemical features is not predicted by a simple one-electron reduction followed by spontaneous anionic polymerization, the two do not exclude one another. For example, it is possible that the resulting polymer can be further reduced. However, there is simply not enough evidence to know the true mechanism of reductive VC polymerization in a cell. The IMC heat flow provides new information that may yet prove useful, but its interpretation is not trivial. Therefore, further theoretical studies should endeavour to determine the origin of the two electrochemical features or, alternatively, propose alternate reaction pathways. These efforts may benefit greatly by comparing the predicted heat flows with the measured amount associated with SEI formation.

The electrochemical and calorimetric plots from FEC-containing cells each display three well-resolved peaks (Figure 3c). From Figure 2a, the cell voltage for these features is approximately 2.4 V, 2.6 V and 2.8 V, and the corresponding maximum heat flow is 4.48 mW (2.24 V), 4.30 mW (2.15 V),

and 3.56 mW (1.78 V), respectively. The heat flow of the first two features is much greater than observed for VC, whereas the cell voltage and maximum heat flow of the third peak is quite similar to that observed for the latter peak in the VC plot (Figure 3b). Therefore, it is possible that some of the FEC is indeed converted into VC and subsequently polymerizes.^{49,61} However, the exact interpretation of these results is not obvious and merits a dedicated investigation to elucidate the electrochemical and chemical processes.

Next, the effect of changing the amount of an additive was examined. This example will also demonstrate exactly how IMC may be used to test possible SEI formation pathways. For this purpose, the SEI formation was measured in cells prepared with 2% or 4% PBF (Figures 3d – e). The coulombic charge and the associated amount of heat associated with the first peak (as estimated from the area of the heat flow plot) varies with the concentration of PBF added to the cell. However, the second peak, previously attributed to the reduction of EC,⁴¹ is not significantly affected, apart from a shift to higher Q . Indeed, comparison with the heat flow of the control cell (Figure 3a) reveals that the reduction of PBF has little or no impact on the reduction of EC at the graphite surface.

A possible reaction pathway for SEI formation in PBF-containing cells was recently reported.²⁵ The authors proposed a pathway that may be simplified and written as two electrochemical steps:



The full rationale for this scheme, including experimental evidence against several alternate pathways is provided in the original article.²⁵ From this simplified scheme, the standard potential was calculated for each step, $E_1^0 = 1.27$ V vs Li/Li⁺ and $E_2^0 = 3.92$ V vs. Li/Li⁺, using density functional theory (DFT).²⁵ If the initial potential of the positive electrode is estimated at ~ 3.5 V vs Li/Li⁺, then this scheme predicts a single voltage plateau during SEI formation that corresponds to a two-electron process and that occurs at $E_{\text{cell}} \approx 2.2$ V. This is in good agreement with the observed voltage plateaus in Figure 2a, from cells that contained 2% or 4% PBF. Furthermore, because the second reduction step occurs at an overpotential, it is predicted that heat will be released during this reaction. In general, the heat flow from an electrochemical reaction may be calculated from the product of the applied current and the overpotential:

$$\left(\frac{dq}{dt}\right)_{\text{chem}} = I \left(\frac{\eta_1 + \eta_2}{n}\right) \quad (3)$$

In this equation, I is the applied current (2 mA in this work), η_1 and η_2 are the overpotentials for each reduction step, and $n = 2$, the overall number of electrons in the pathway. For the

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small current density used in the present work, the approximation is made that the negative electrode is near the first equilibrium potential and the heat flow for SEI formation is calculated:

$$\eta = \frac{(E_1^0 - E) + (E_2^0 - E)}{2} \quad (4)$$

$$\left(\frac{dq}{dt}\right)_{PBF} = (2 \text{ mA}) \cdot \frac{(1.27 \text{ V} - 1.27 \text{ V}) + (3.92 \text{ V} - 1.27 \text{ V})}{2} \quad (5)$$

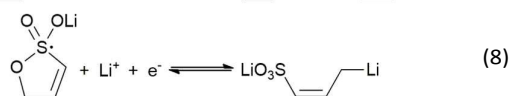
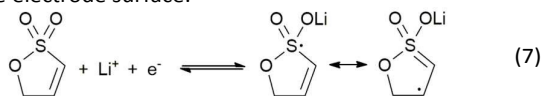
$$= 2.65 \text{ mW}$$

A comparison of this value with Figure 3d-e reveals that this is a slight underestimate of the experimentally measured heat flow. This is because the SEI formation is only occurring at the negative electrode surface, whereas there is an equivalent amount of delithiation occurring at the positive electrode. The heat flow from this process has been characterized previously and is represented by the entropy term within the mean-field approximation:⁴⁴

$$\left(\frac{dq}{dt}\right)_{\text{entropy}} = I \frac{k_B T}{e} \ln\left(\frac{x}{1-x}\right) \quad (6)$$

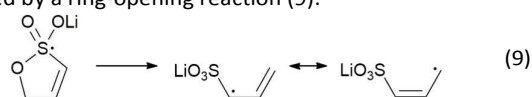
where k_B is the Boltzmann constant, T is the temperature, e is the elementary charge, and x is the degree of lithiation, such that $x = 1$ for the fully lithiated positive electrode. The sum of the heat flow from equations (5) and (6) was then plotted against the experimentally measured heat flow of SEI formation (Figure 4). The capacity for PBF reduction was determined assuming a two-electron process and from the amount of additive introduced into the cell. The result shown in Figure 4 demonstrates that there is indeed very good agreement between the predicted and observed heat flow in these cells, once the entropy contribution from the positive electrode is included.

The final example in this article will test how IMC may be used to evaluate SEI formation pathways involving both electrochemical and chemical steps. For this purpose, the recently reported SEI formation pathway for PES was examined.³⁰ The proposed pathway begins with two consecutive one-electron reduction steps, which produces an organolithium compound, $\text{LiO}_3\text{SCHCH}_2\text{Li}$ (Li_2PES), at the negative electrode surface:

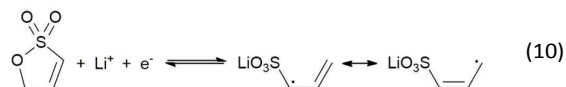


The standard potentials for each electrochemical step have been evaluated by DFT (B3LYP/6-311++G(d,p)/IEFPCM-UFF, $\epsilon = 20$), as $E_7^0 = 0.92 \text{ V}$ vs. Li/Li^+ and $E_8^0 = 4.33 \text{ V}$ vs. Li/Li^+ .^{28,30} It is noted that the cell voltage in the present work ($E_{\text{cell}} \sim 2.2 \text{ V}$, $\therefore E_{\text{neg}} \sim 1.3 \text{ V}$ vs. Li/Li^+) is not an exact match with the experimental value reported by Self *et al.* ($E_{\text{cell}} \sim 2.4 \text{ V}$, $\therefore E_{\text{neg}} \sim 1.1 \text{ V}$ vs. Li/Li^+), nor with their calculated value of PES reduction ($E_7^0 = 0.9 \text{ V}$ vs. Li/Li^+).³⁰ This discrepancy is likely a combination of small differences in the potential of the positive electrode materials (NMC442 was used in the present work, NMC111 was used in Self *et al.*) and the inaccuracy of the DFT calculations. The accuracy of calculated potentials is

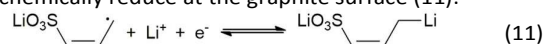
limited by the accuracy of the Li/Li^+ reference value, the accuracy of solvation model used, the size of the basis set, any surface interactions of the oxidized and/or reduced species, and the accuracy of the density functional itself. Nevertheless, the value of the reference potential (Li/Li^+) and, to some extent, the solvation energies cancel out when calculated potentials are used to predict heat flow. Moreover, these two calculated values, i.e., E_1^0 and E_2^0 , provide the valuable insight that a) the observed dQ/dV peaks in this work and in the work of Self *et al.* may be attributed to the one-electron reduction of PES in solution, and b) the subsequent reduction step is spontaneous. This second point leads to a discussion of the thermodynamic predictions of DFT calculations and the kinetic realities of complex chemical systems. When reviewing the above reaction scheme, one might alternatively suggest that the initial electrochemical reduction of PES (7) is instead followed by a ring-opening reaction (9):



The calculated free energy change of reaction (9) is -179 kJ mol^{-1} (-1.86 eV) and the enthalpy change is -177 kJ mol^{-1} (-1.84 eV). One might also suggest that ring-opening could occur as part of the initial reduction of PES (10):⁵⁴



The product of reactions (9) and (10) are the same, but they may be distinguished by the electrochemical potential of their corresponding reduction steps. The standard potential of the two-step mechanism (reduction followed by ring-opening) is determined by the electrochemical step and is therefore the same as the pathway reported by Self *et al.* and discussed above ($E_7^0 = 0.9 \text{ V}$ vs. Li/Li^+). In contrast, the calculated potential of the one-step process is $E_{10}^0 = 2.78 \text{ V}$ vs. Li/Li^+ (cf. $E_{\text{exp}} = 1.1 - 1.3 \text{ V}$ vs. Li/Li^+).³⁰ Whereas the poor comparison of DFT and experimental potentials suggest that reaction (10) does not occur, reaction (9) remains a possibility. Its product could then electrochemically reduce at the graphite surface (11):



Thus, there are two possible pathways for the two-electron reduction of PES, reactions (7) and (8) or reactions (7), (9), and (11). Although the final product is the same, the standard potentials of the second reduction step is not. Using the same DFT method as Self *et al.*, the calculated standard potentials for reactions (8) and (11) are $E_8^0 = 4.33 \text{ V}$ vs. Li/Li^+ and $E_{11}^0 = 2.48 \text{ V}$ vs. Li/Li^+ , respectively. Both reactions are spontaneous at the electrode potential of the first reduction step (because $E_8^0 > E_7^0$ and $E_{11}^0 > E_7^0$) and would both result in a single two-electron reduction feature. That is, electrochemical measurements cannot differentiate these two reaction pathways and the correct mechanism remains unknown.

The two pathways discussed above result in the same reduced species, Li_2PES . If one's goal is to learn about the chemical

composition of the SEI, then the exact mechanism may seem unnecessary. However, such detail may yet prove essential to determine the final product or products of PES reduction. Nevertheless, the subsequent reactions proposed by Self *et al.* may next be considered. The authors suggested that the reduced Li_2PES species reacts with nearby solvent molecules and/or non-reduced PES to form the SEI.³⁰ Eighteen possible chemical reactions were proposed, the products of which are consistent with XPS surface analysis, but are otherwise difficult to confirm or deny. By evaluating their spontaneity, one might immediately eliminate some reactions. In this case, DFT has been used to determine that all eighteen reactions are indeed spontaneous.³⁰ However, this is not to suggest that all of these reactions actually occur in a cell. Rather, the resulting list of possibilities is just that, a summary of hypothetical SEI formation reactions. Whereas reaction energies are readily calculated using DFT, reaction kinetics are not so easily predicted and there is no substitute for good experimental studies. Therefore, one goal of the present work is to assess whether IMC may be used to identify which of these reactions actually occur or, at least, to narrow down the large list of possibilities. Because Self *et al.* calculated thermodynamic values for each of these chemical reactions,³⁰ the overall heat flow of SEI formation in PES-containing cells may be calculated. The heat flow from the electrochemical steps is the same as discussed for PBF reduction above. For a chemical reaction, the heat flow may be determined from the enthalpy of reaction. Therefore, the heat flow from the proposed reaction pathway may be written as:

$$\left(\frac{dq}{dt}\right)_{\text{calc}} = -I \left(\frac{\eta_1 + \eta_2}{n} + \frac{\Delta H}{nF} \right) \quad (12)$$

In this equation, I is the applied current (2 mA in this work), η_1 and η_2 are the overpotentials for each reduction step, ΔH is the enthalpy of the chemical reaction step, F is Faraday's constant, and $n = 2$, the overall number of electrons in the pathway. As previously, the approximation is made that at low current density, the negative electrode is near the first equilibrium potential:

$$\begin{aligned} \eta_1 &= E_1^0 - E \\ &\approx 0 \text{ V} \end{aligned}$$

Therefore, the overpotential of the second reduction step may also be evaluated:

$$\begin{aligned} \eta_2 &= E_2^0 - E \\ &\approx (4.3 \text{ V}) - (0.9 \text{ V}) \\ &\approx 3.4 \text{ V} \end{aligned}$$

It is noteworthy that the absolute values of the potentials are not important for this calculation. Rather, it is the potential of the second reduction step, relative to the first that affects the heat flow. Therefore, any systematic calculation error due to effects such as Li^+ cation solvation will cancel.

Finally, the reaction of Li_2PES with PES, EC, or EMC may be considered. Self *et al.* proposed eighteen possible reaction products (Figure S1) and calculated the free energy change for each (Figure 8 in Self *et al.*)³⁰ Although the original article only reported electronic energies and free energies, the enthalpy values are automatically calculated when obtaining free

energy values. Therefore, these results were obtained from the authors and are included as Supplementary Material with this article, including a comparison of ΔH and ΔG for all 18 reactions (Figure S2). Future DFT and *ab initio* studies are encouraged to also report the enthalpy change for comparison with IMC measurements. These values are all exothermic and are spread over the range $-0.046 \text{ eV} < \Delta H < -3.67 \text{ eV}$. Based on the enthalpy change, the 18 reactions were placed into four groups: Reactions with EC or EMC that yield lithium alkoxides (-0.046 to -0.63 eV), reactions with EMC that produce lithium semicarbonates (-1.68 to -1.98 eV), reactions with PES (-2.21 to -2.57 eV), and reactions with EC that produce lithium semicarbonates (-3.63 to -3.67 eV). Given the kinetic parameters are unknown, it is unknown how likely these reactions are to occur. For the purpose of predicting the heat flow during SEI formation, the mean enthalpy change for each category was calculated: -0.3 eV , -1.8 eV , -2.3 eV , and -3.6 eV , respectively. Note that because these energies are reported in molar electron-volts, the value of Faraday's constant, F is one. Therefore, the calculated heat flows from equation (12) are 3.7 mW, 5.2 mW, and 5.7 mW, and 7.0 mW, respectively. It is noteworthy that the alternate reduction pathway, which involves a ring-opening step prior to the second electrochemical reduction, gives the same predicted heat flows, within 0.02 mW. Because the heat flow is essentially unaffected by the reduction pathway, IMC cannot be applied to study this part of the reaction mechanism. Conversely, this part of the reaction mechanism does not interfere with the ability of IMC to study the chemical reactions that occur following the electrochemical reduction of PES.

The total heat flow during SEI formation in PES-containing cells, calculated from the sum of equations (6) and (12), was then plotted with the experimental data (Figure 5a). It was approximated that PES reduction stops after 3.5 mA h (13 C) of charge has passed, as estimated from the dQ/dV vs. Q plot. For comparison, the charge associated with the two-electron reduction of all the PES originally in the cell (0.018 g) would be 8.0 mA h (29 C). This observation suggests that the PES passivation layer reaches some maximum thickness that cannot be predicted by any simple approach. The results show that the experimental heat flow is within the range of predicted values, although it is not a close match to any of the four groups outlined above. This supports the idea that several or even all of the eighteen final products are formed.

One notable difference between PBF- and PES-containing cells is the sharpness of the initial rise in heat flow as SEI formation first begins. For PBF, the heat flow quickly increases until it nears the predicted line shape. The difference between the calculated curve and the experimentally measured heat flow is thought to predominantly originate from the time constant of the calorimeter ($\tau = 5 \text{ min}$). Whereas in PES-containing cells, the heat flow approaches the calculated value much more gradually (Figure 5b). The times for the heat flow to reach one-half and three-quarters of the maximum value are 11 min and 25 min, respectively, for an apparent time constant of $\sim 12 \text{ min}$. This difference may be attributable to kinetic differences between the two reaction pathways; the reduction of PBF

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involves chemical reactions of radical species, which may be presumed to occur very rapidly. In contrast, the reduction of PES involves two electrochemical steps followed by chemical reactions involving an organolithium compound, Li_2PES . Therefore, the longer apparent time constant in PES-containing cells is here attributed to the kinetics of this chemical reaction of Li_2PES . Alternatively, it is possible that this should be attributed to the kinetics of the ring-opening reaction (9). Given the importance of kinetic factors in determining which pathway SEI formation follows and the limitations of DFT to predict these factors, a new technique for kinetic studies could be quite valuable. Therefore, future studies may consider how IMC may be applied for *in situ* chemical kinetic studies during SEI formation.

Conclusions

This article demonstrates how IMC may be used in conjunction with electrochemical measurements to study the electrochemical and chemical processes that occur during SEI formation in lithium-ion cells. The heat flow was measured for SEI formation in control cells, which contained no electrolyte additives, and from cells prepared with four additives that are known to produce a passive SEI at the negative electrode surface: VC, FEC, PBF, and PES. Plotting the differential capacity (dQ/dV) and heat flow (dq/dt) against the cell capacity (Q) allows easy visualization of fine details during SEI formation. For VC, there are two distinct dQ/dV features (2.65 V, 2.8 V), which align with a broad peak in the IMC plot (3.78 mW). The results for FEC show three distinct dQ/dV (2.4 V, 2.6 V, 2.8V) and calorimetric (4.48 mW/2.24 V, 4.30 mW/2.15 V, 3.56 mW/1.78 V) peaks. It is demonstrated that repeated measurements at multiple concentrations may provide new insight into SEI formation in PBF-containing cells, where the reduction of PBF does not significantly affect the reduction of EC at higher cell voltage. Finally, it is demonstrated how IMC measurements may be used in conjunction with DFT calculations to assess proposed reaction pathways. The measured heat flow for PBF- and PES-containing cells matches the predicted ranges for the reaction pathways proposed by Hall *et al.* and Self *et al.*^{25,30} The possibility of applying this method to obtain kinetic information could be quite valuable but requires further investigation. It is hoped that these general methods will be adopted and refined to study the effects of electrolyte additives on the underlying chemical mechanisms of SEI formation in lithium-ion cells.

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References

- 1 K. Xu, *Chem. Rev.*, 2014, **114**, 11503–11618.
- 2 H. Bang, H. Yang, Y. K. Sun and J. Prakash, *J. Electrochem. Soc.*, 2005, **152**, A421–A428.
- 3 H. Joachin, T. D. Kaun, K. Zaghib and J. Prakash, *J. Electrochem. Soc.*, 2009, **156**, A401–A406.
- 4 J.-S. Kim, J. Prakash and J. R. Selman, *Electrochem. Solid-State Lett.*, 2001, **4**, A141–A144.
- 5 W. Lu, I. Belharouak, J. Liu and K. Amine, *J. Power Sources*, 2007, **174**, 673–677.
- 6 W. Lu, I. Belharouak, D. Vissers and K. Amine, *J. Electrochem. Soc.*, 2006, **153**, A2147–A2151.
- 7 W. Lu, I. Belharouak, S. H. Park, Y. K. Sun and K. Amine, *Electrochimica Acta*, 2007, **52**, 5837–5842.
- 8 W. R. McKinnon, J. R. Dahn, J. J. Murray, R. R. Haering, R. S. McMillan and A. H. Rivers-Bowerman, *J. Phys. C Solid State Phys.*, 1986, **19**, 5135.
- 9 Y. Saito, K. Takano, K. Kanari, A. Negishi, K. Nozaki and K. Kato, *J. Power Sources*, 2001, **97–98**, 688–692.
- 10 K. E. Thomas and J. Newman, *J. Electrochem. Soc.*, 2003, **150**, A176–A192.
- 11 H. Yang and J. Prakash, *J. Electrochem. Soc.*, 2004, **151**, A1222–A1229.
- 12 W. Lu, I. Belharouak, J. Liu and K. Amine, *J. Electrochem. Soc.*, 2007, **154**, A114–A118.
- 13 W. Lu and J. Prakash, *J. Electrochem. Soc.*, 2003, **150**, A262–A266.
- 14 J.-S. Hong, H. Maleki, S. A. Hallaj, L. Redey and J. R. Selman, *J. Electrochem. Soc.*, 1998, **145**, 1489–1501.
- 15 Y. Kobayashi, H. Miyashiro, K. Kumai, K. Takei, T. Iwahori and I. Uchida, *J. Electrochem. Soc.*, 2002, **149**, A978–A982.
- 16 Y. Saito, K. Kanari and K. Takano, *J. Power Sources*, 1997, **68**, 451–454.
- 17 Y. Saito, M. Shikano and H. Kobayashi, *J. Power Sources*, 2013, **244**, 294–299.
- 18 L. E. Downie, L. J. Krause, J. C. Burns, L. D. Jensen, V. L. Chevrier and J. R. Dahn, *J. Electrochem. Soc.*, 2013, **160**, A588–A594.
- 19 L. E. Downie and J. R. Dahn, *J. Electrochem. Soc.*, 2014, **161**, A1782–A1787.
- 20 L. J. Krause, L. D. Jensen and J. R. Dahn, *J. Electrochem. Soc.*, 2012, **159**, A937–A943.
- 21 V. L. Chevrier, L. Liu, D. B. Le, J. Lund, B. Molla, K. Reimer, L. J. Krause, L. D. Jensen, E. Figgemeier and K. W. Eberman, *J. Electrochem. Soc.*, 2014, **161**, A783–A791.
- 22 L. E. Downie, K. J. Nelson, R. Petibon, V. L. Chevrier and J. R. Dahn, *ECS Electrochem. Lett.*, 2013, **2**, A106–A109.
- 23 L. E. Downie, S. R. Hyatt, A. T. B. Wright and J. R. Dahn, *J. Phys. Chem. C*, 2014, **118**, 29533–29541.
- 24 L. E. Downie, Ph.D. Chemistry, Dalhousie University, 2015.
- 25 D. S. Hall, M. Nie, L. Ellis, S. L. Glazier, S. Hyatt, R. Petibon, A. Xiao, W. M. Lamanna, K. Smith, I. G. Hill and J. R. Dahn, *J. Electrochem. Soc.*, 2016, **163**, A773–A780.
- 26 O. Borodin and D. Bedrov, *J. Phys. Chem. C*, 2014, **118**, 18362–18371.
- 27 O. Borodin and G. D. Smith, *J. Phys. Chem. B*, 2009, **113**, 1763–1776.
- 28 D. S. Hall, J. Self and J. R. Dahn, *J. Phys. Chem. C*, 2015, **119**, 22322–22330.
- 29 M. T. Ong, O. Verners, E. W. Draeger, A. C. T. van Duin, V. Lordi and J. E. Pask, *J. Phys. Chem. B*, 2015, **119**, 1535–1545.

- 30 J. Self, D. S. Hall, L. Madec and J. R. Dahn, *J. Power Sources*, 2015, **298**, 369–378.
- 31 N. Takenaka, H. Sakai, Y. Suzuki, P. Uppala and M. Nagaoka, *J. Phys. Chem. C*, 2015, **119**, 18046–18055.
- 32 N. Takenaka, Y. Suzuki, H. Sakai and M. Nagaoka, *J. Phys. Chem. C*, 2014, **118**, 10874–10882.
- 33 K. Ushirogata, K. Sodeyama, Y. Okuno and Y. Tateyama, *J. Am. Chem. Soc.*, 2013, **135**, 11967–11974.
- 34 K. Ushirogata, K. Sodeyama, Z. Futera, Y. Tateyama and Y. Okuno, *J. Electrochem. Soc.*, 2015, **162**, A2670–A2678.
- 35 A. von Wald Cresce, O. Borodin and K. Xu, *J. Phys. Chem. C*, 2012, **116**, 26111–26117.
- 36 A. von Wald Cresce, M. Gobet, O. Borodin, J. Peng, S. M. Russell, E. Wikner, A. Fu, L. Hu, H.-S. Lee, Z. Zhang, X.-Q. Yang, S. Greenbaum, K. Amine and K. Xu, *J. Phys. Chem. C*, 2015, **119**, 27255–27264.
- 37 D. Aurbach, K. Gamolsky, B. Markovsky, Y. Gofer, M. Schmidt and U. Heider, *Electrochimica Acta*, 2002, **47**, 1423–1439.
- 38 J. C. Burns, R. Petibon, K. J. Nelson, N. N. Sinha, A. Kassam, B. M. Way and J. R. Dahn, *J. Electrochem. Soc.*, 2013, **160**, A1668–A1674.
- 39 L. El Ouatani, R. Dedryvère, C. Siret, P. Biensan, S. Reynaud, P. Iratçabal and D. Gonbeau, *J. Electrochem. Soc.*, 2009, **156**, A103–A113.
- 40 I. A. Shkrob, J. F. Wishart and D. P. Abraham, *J. Phys. Chem. C*, 2015, **119**, 14954–14964.
- 41 M. Nie, J. Xia and J. R. Dahn, *J. Electrochem. Soc.*, 2015, **162**, A1186–A1195.
- 42 M. Nie, D. Chalasani, D. P. Abraham, Y. Chen, A. Bose and B. L. Lucht, *J. Phys. Chem. C*, 2013, **117**, 1257–1267.
- 43 K. Xu, *Chem. Rev.*, 2004, **104**, 4303–4418.
- 44 J. R. Dahn, W. R. McKinnon, J. J. Murray, R. R. Haering, R. S. McMillan and A. H. Rivers-Bowerman, *Phys. Rev. B*, 1985, **32**, 3316–3318.
- 45 M. Broussely, P. Biensan, F. Bonhomme, P. Blanchard, S. Herreyre, K. Nechev and R. J. Staniewicz, *J. Power Sources*, 2005, **146**, 90–96.
- 46 L. El Ouatani, R. Dedryvère, C. Siret, P. Biensan, S. Reynaud, P. Iratçabal and D. Gonbeau, *J. Electrochem. Soc.*, 2009, **156**, A103.
- 47 L. Ma, J. Xia and J. R. Dahn, *J. Electrochem. Soc.*, 2015, **162**, A1170–A1174.
- 48 J. Xia, L. Ma, C. P. Aiken, K. J. Nelson, L. P. Chen and J. R. Dahn, *J. Electrochem. Soc.*, 2014, **161**, A1634–A1641.
- 49 V. Etacheri, O. Haik, Y. Goffer, G. A. Roberts, I. C. Stefan, R. Fasching and D. Aurbach, *Langmuir*, 2012, **28**, 965–976.
- 50 E. Markevich, G. Salitra, K. Fridman, R. Sharabi, G. Gershinsky, A. Garsuch, G. Semrau, M. A. Schmidt and D. Aurbach, *Langmuir*, 2014, **30**, 7414–7424.
- 51 M.-H. Ryou, G.-B. Han, Y. M. Lee, J.-N. Lee, D. J. Lee, Y. O. Yoon and J.-K. Park, *Electrochimica Acta*, 2010, **55**, 2073–2077.
- 52 B. Li, M. Xu, T. Li, W. Li and S. Hu, *Electrochem. Commun.*, 2012, **17**, 92–95.
- 53 B. Li, Y. Wang, H. Rong, Y. Wang, J. Liu, L. Xing, M. Xu and W. Li, *J. Mater. Chem. A*, 2013, **1**, 12954.
- 54 B. Li, M. Xu, B. Li, Y. Liu, L. Yang, W. Li and S. Hu, *Electrochimica Acta*, 2013, **105**, 1–6.
- 55 B. Li, Y. Wang, W. Tu, Z. Wang, M. Xu, L. Xing and W. Li, *Electrochimica Acta*, 2014, **147**, 636–642.
- 56 M. Nie, L. Madec, J. Xia and J. R. Dahn, 2015.
- 57 L. El Ouatani, R. Dedryvère, C. Siret, P. Biensan and D. Gonbeau, *J. Electrochem. Soc.*, 2009, **156**, A468–A477.
- 58 H. Ota, Y. Sakata, A. Inoue and S. Yamaguchi, *J. Electrochem. Soc.*, 2004, **151**, A1659–A1669.
- 59 B. Zhang, M. Metzger, S. Solchenbach, M. Payne, S. Meini, H. A. Gasteiger, A. Garsuch and B. L. Lucht, *J. Phys. Chem. C*, 2015.
- 60 J. Self, M.Sc. Physics, Dalhousie University, 2015, <http://hdl.handle.net/10222/60315>.
- 61 X. Zhao, Q.-C. Zhuang, S.-D. Xu, Y.-X. Xu, Y.-L. Shi and X.-X. Zhang, *Int J Electrochem Sci*, 2015, **10**, 2515–2534.
- 62 X. Chen, X. Li, D. Mei, J. Feng, M. Y. Hu, J. Hu, M. Engelhard, J. Zheng, W. Xu, J. Xiao, J. Liu and J.-G. Zhang, *ChemSusChem*, 2014, **7**, 549–554.
- 63 K. Leung, S. B. Rempe, M. E. Foster, Y. Ma, J. M. M. del la Hoz, N. Sai and P. B. Balbuenaa, *J. Electrochem. Soc.*, 2014, **161**, A213–A221.
- 64 H. Nakai, T. Kubota, A. Kita and A. Kawashima, *J. Electrochem. Soc.*, 2011, **158**, A798–A801.

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Figures

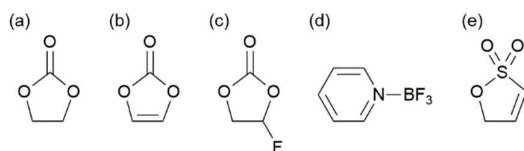


Figure 1 The molecular structures of the SEI-forming compounds examined in this work: a) ethylene carbonate (EC), b) vinylene carbonate (VC), c) fluoroethylene carbonate (FEC), d) pyridine boron trifluoride (PBF), and e) prop-1-ene-1,3-sultone (PES).

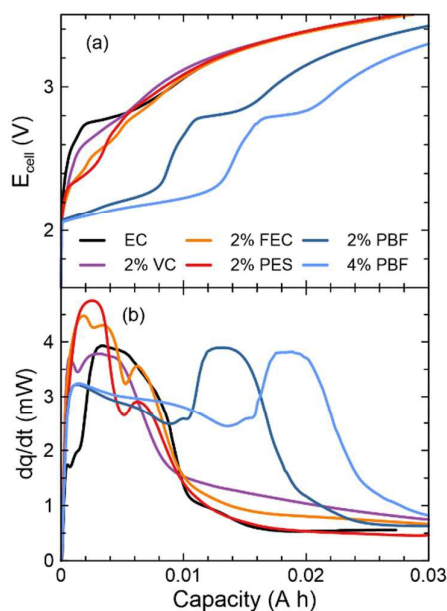


Figure 2 The a) cell voltage and b) heat flow were measured during the initial charge at 2 mA ($\sim C/10$) from NMC442/graphite cells that contained control electrolyte (3:7 EC/EMC, 1 mol L⁻¹ LiPF₆) and various additives (as indicated on the figure).

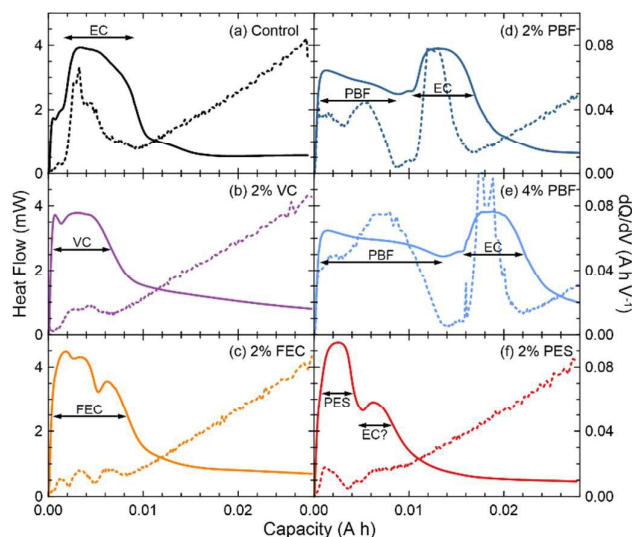


Figure 3 The heat flow and differential capacity were measured during the initial charge at 2 mA ($\sim C/10$) from NMC442/graphite cells that contained a) control electrolyte (3:7 EC/EMC, 1 mol L⁻¹ LiPF₆) and b) 2% VC, c) 2% FEC, d) 2% PBF, e) 4% PBF, or f) 2% PES. Solid lines are heat flow, dashed lines are dQ/dV .

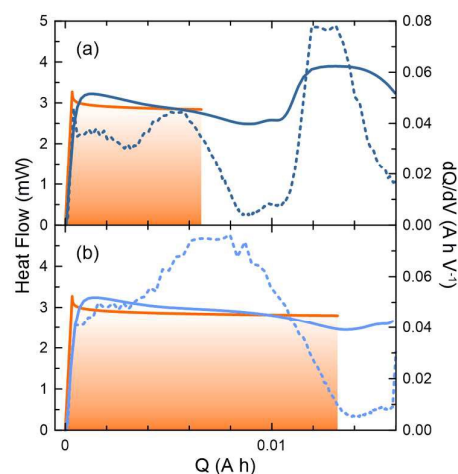


Figure 4 The experimental heat flow during SEI formation in cells containing a) 2% PBF and b) 4% PBF is compared with the predicted heat flow from equation (5) and the entropy contribution from equation (6).³⁰ Solid lines are heat flow, dashed lines are dQ/dV .

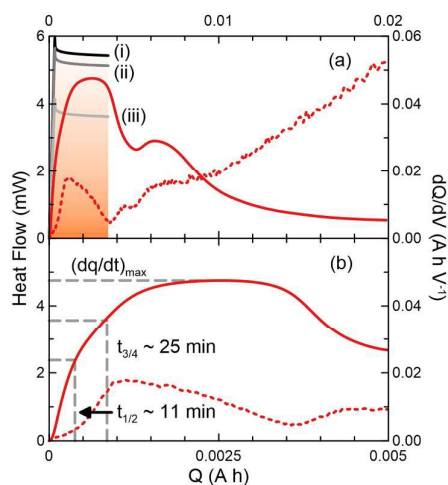


Figure 5 a) The experimental heat flow during SEI formation in a PES-containing cell is compared with the predicted heat flow. The heat flow from delithiation of the positive electrode, calculated from equation (6), was added to the heat flow of SEI formation, i) 7.0 mW, ii) 5.7 mW, iii) 5.2 mW, and iv) 3.7 mW, as calculated from equation (12) and the DFT results of Self et al.³⁰ Solid lines are heat flow and dotted lines are dQ/dV . b) The time for the heat flow to reach one half ($t_{1/2}$) and three quarters ($t_{3/4}$) of the maximum value are indicated by the dashed lines.